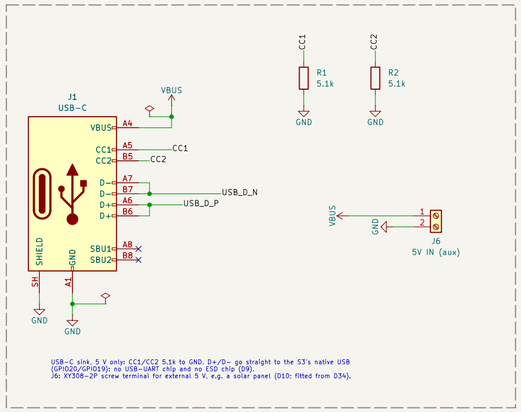
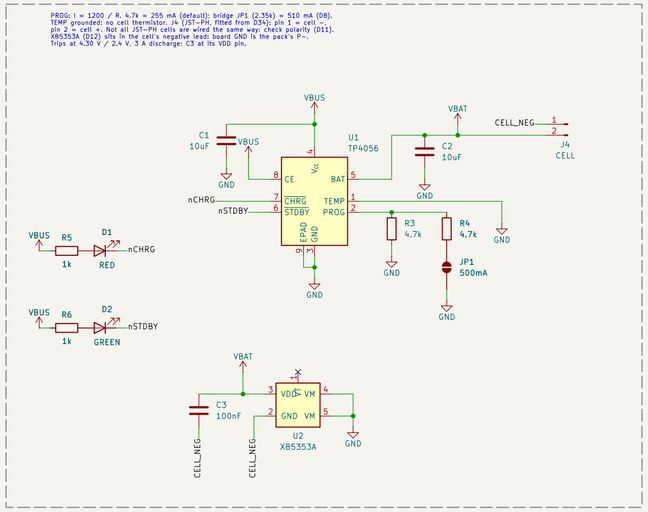
USB-C: 5 V IN AND PROGRAMMING

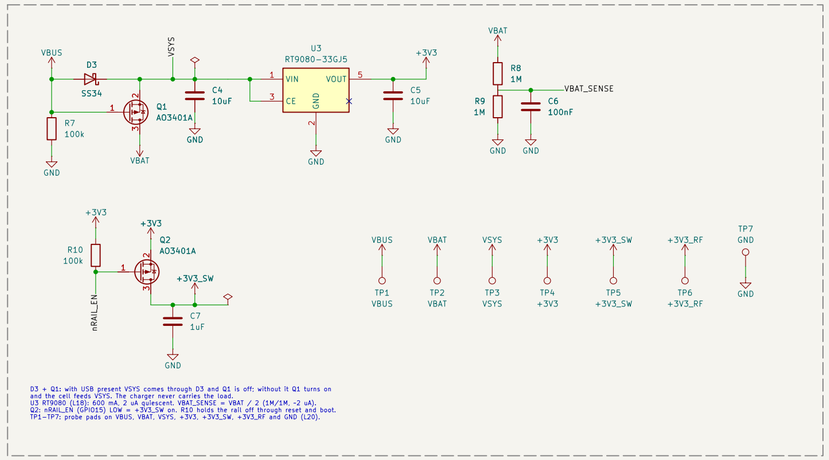


USB-C sink: 5 V only: CC1/CC2 5.1k to GND. D+/D− go straight to the S3's native USB (GPIO20/GPIO19); no USB-UART chip and no ESD chip (D9).
J6: XY308-2P screw terminal for external 5 V, e.g. a solar panel (D10; fitted from D34).

LI-ION CHARGER & CELL PROTECTION

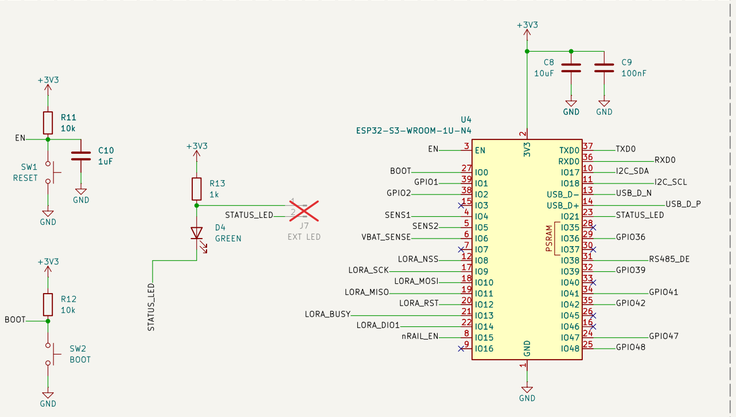


SYSTEM POWER: LOAD SHARING, 3.3 V LDO, SWITCHED SENSOR RAIL



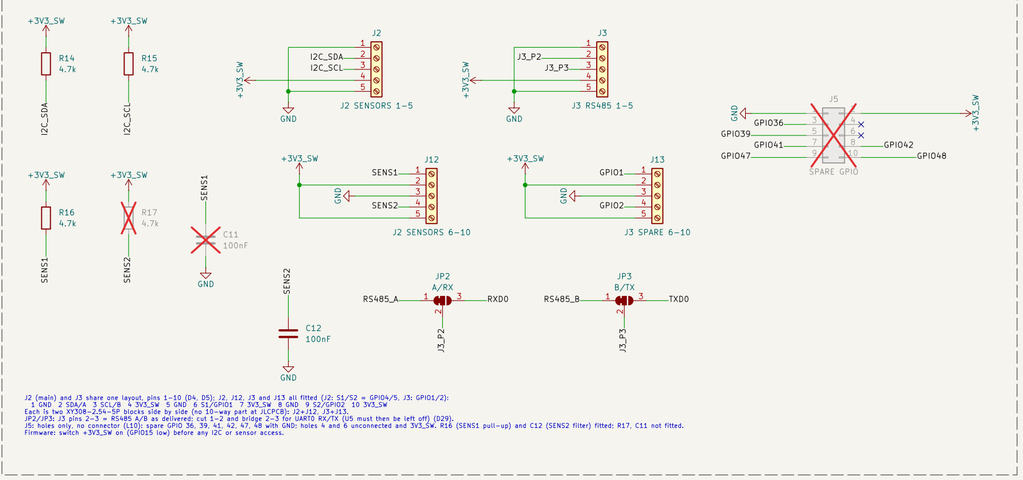
D3 + Q1: with USB present VSYS comes through D3 and Q1 is off; without it Q1 turns on and the cell feeds VSYS. The charger never carries the load.
U3 RT9080 (L18): 600 mA, 2 uA quiescent. VBAT_SENSE = VBAT / 2 (1M/1M, ~2 uA).
Q2: nRAIL_EN (GPIO15) LOW = +3V3_SW on. R10 holds the rail off through reset and boot.
TP1–TP7: probe pads on VBUS, VBAT, VSYS, +3V3, +3V3_SW, +3V3_RF and GND (L20).

ESP32-S3-WROOM-1U MODULE



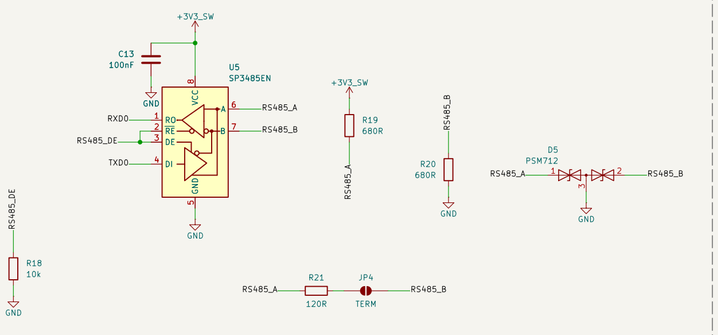
GPIO map (L3, L19): sensors 4/5 (ADC1) -> J2, spares 1/2 (ADC1) -> J3, VBAT_SENSE 6, I2C 17/18, LoRa NSS/SCK/MOSI/MISO 8–11, RST 12, BUSY 13, DIO1 14 (as Heltec V3/V4), nRAIL_EN 15, RS485 DE 36 (UART0 43/44 = RS485 or J3 USB 19/20, LED 21, BOOT 0.
Spares 36, 39, 41, 42, 47, 48 -> J5 holes (L10, L19). Not connected: straps 3, 45, 46 (internal defaults) and 7, 16, 35, 37, 40. J7 (not fitted): off-board LED, pin 1 = anode, pin 2 = cathode, shares R13.
C8/C9 close to module pin 2; U.FL antenna: module end at the board edge, away from the LoRa antenna (L5).

SENSOR CONNECTORS J2, J3 (FITTED) AND SPARE GPIO HOLES J5



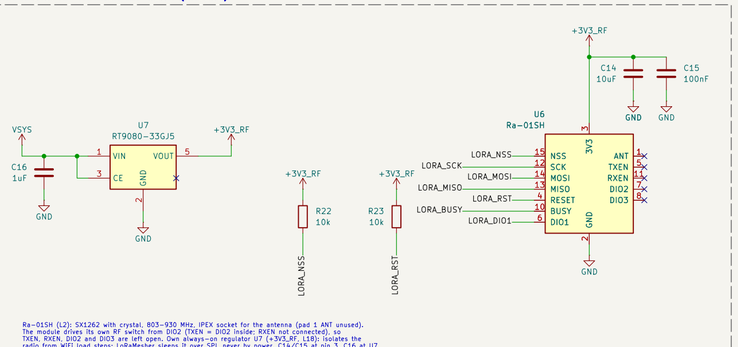
J2 (main) and J3 share one layout, pins 1–10 (D4, D5): J2, J12, J3 and J13 all fitted (J2: S1/S2 = GPIO4/5, J3: GPIO1/2):
1 GND 2 SDA/A 3 SCL/B 4 3V3_SW 5 GND 6 S1/GPIO1 7 3V3_SW 8 GND 9 S2/GPIO2 10 3V3_SW
Each is two XH308-2J4-5P blocks side by side (no 10-way part at JLCPCB): J2+J12, J3+J13.
JP2/JP3: J3 pins 2–3 = RS485 A/B as delivered; cut 1–2 and bridge 2–3 for UART0 RX/TX (U5 must then be left off) (D29).
J5: holes only, no connector (L10); spare GPIO 36, 39, 41, 42, 47, 48 with GND: holes 4 and 6 unconnected and 3V3_SW. R16 (SENS1 pull-up) and C12 (SENS2 filter) fitted: R17, C11 not fitted.
Firmware: switch +3V3_SW on (GPIO15 low) before any I2C or sensor access.

RS485 / MODBUS (FITTED; OMIT TO SAVE COST)



Fitted by default: J3 pin 2 = A, pin 3 = B via JP2/JP3 (D29). A board without Modbus can leave out U5, C13, R18–R21, D5 and J3 (see COST-ESTIMATE.txt). GPIO38 = DE/RE (high = transmit, R18 = receive).
JP4 bridged = 120R termination. RS485 sensors need their own 5–24 V supply (external).

LORA RADIO: AI-THINKER RA-01SH (SX1262)



Ra-01SH (L2): SX1262 with crystal, 803–930 MHz. IPEX socket for the antenna (pad 1 ANT unused).
The module drives its own RF switch from DIO2 (TXEN = DIO2 inside: RXEN not connected), so TXEN, RXEN, DIO2 and DIO3 are left open. Own always-on regulator U7 (+3V3_RF, L18): isolates the radio from WiFi load steps; LoRaMesher sleeps it over SPI, never by power. C14/C15 at pin 3, C16 at U7. R22 keeps the radio deselected and R23 keeps it out of reset while the ESP32 boots or sleeps.
Firmware: LoRaMesher/RadioLib defaults fit (DC-DC regulator, no TCXO, DIO2 as RF switch) (L6).

Generated by claude-tools/schematic/draw.py from circuit.py -- edit those, not this file

Natural Innovation

Sheet: /

File: frugal_iot_v03.kicad_sch

Title: Frugal-IoT V0.3 LoRa

Size: A2	Date: 2026-10-31	**Rev: 0.3**
KiCad E.D.A. 10.0.4		Id: 1/1